

Vaibhav Balloli

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Research Statement

Naively deployed AI systems fail because underlying capabilities are not reliably translated for the consumers. I conduct applied research that investigate how to build agents tailored towards deployment needs across domains like healthcare and conservation. My work addresses four parts of that gap:

1. Post-training environments for agents to clarify intent, recover, abstain, or delegate.
2. Contextual LLM and agent evaluations built with domain experts.
3. Science of understanding and improving instruction-following behaviors and capabilities of agents
4. Monitoring these systems at test time.

My research has always aimed towards building and deploying human-AI interaction-centered approach with stakeholders in challenging domains like reproductive health, wildlife conservation, and road safety - identifying and researching how AI can solve problems, coordinating multi-stakeholder requirements, publishing relevant papers (NeurIPS, COLING, KDD, FAccT, and IJCAI), and deployed them in the real-world.

Research and Engineering Deployments Impact

- **PATHFinder (health)**: Conversational patient- and clinician-facing LLM agent being deployed with clinicians and patients to personalize prenatal care planning at the University of Michigan.
- **Elephant Re-ID (conservation)**: Deploying novel, SOTA computer vision models in Botswana.
- **HAMS (road safety)**: Automated driver's-license testing deployed nationally across India; **440,000+ candidates** assessed, adopted by state transport authorities ([live dashboard](#)).

Technical Skills

- **Languages and frameworks**: Python, C++, JavaScript; PyTorch, JAX, SkyRL, verl.

Education

University of Michigan, Ann Arbor (GPA: 4.0) *2023 - Present (expected Dec 2027)*
Ph.D. Candidate and M.S., Computer Science and Engineering.

BITS Pilani, Hyderabad Campus *2016 - 2020*
B.E., Electronics and Communication Engineering

Research Experience

Microsoft Research Redmond (AI Frontiers), Research Intern *May 2025 – Aug 2025*
Advisors: Hussein Mozannar, Adam Fourney, Gagan Bansal, Saleema Amershi, Eric Horvitz

- Applied multi-turn reinforcement learning (GRPO) to improve how LLM agents clarify intent and recover from errors across extended episodes, and built evaluation harnesses for multi-turn agent performance. Under submission with five Microsoft Research co-authors.

Microsoft Research India, Research Fellow *June 2021 – June 2023*
Research Fellow (2022–23); Societal Impact through Cloud and AI Fellow (2021–22). Advisors: Dr. Akshay Nambi, Tanuja Ganu, Dr. Venkat Padmanabhan

- **VeLLM**: Built runtime strategies that select prompt, LLM, and embedding model per query, plus the multilingual evaluation and retrieval components of the platform. Evaluated across **18 languages: 10–15% gains** over pre-trained and **4x** over fine-tuned language-specific baselines, with no additional training (COLING 2025, oral).
- **HAMS**: Visual SLAM, object detection, and trajectory analysis for automated driver's-license testing.

- **Chanakya**: Contextual-bandit framework learning runtime accuracy/latency tradeoffs for streaming perception, beating static and dynamic baselines on server GPUs and edge devices (NeurIPS 2023).

Publications and Working Papers

9. **Fragile at the Decision Boundary: Evaluating LLM and Decision Model Capabilities**
Vaibhav Balloli, Elizabeth Bondi-Kelly
Under submission
8. **MATTER-LC: Multimodal Agentic Task Orchestration for Transforming Evidence into Source-Auditable Records in Landslide-Loss Curation**
Xin Wei, Sabine Loos, **Vaibhav Balloli**, Lauren K. Palermo, Benjamin B. Mirus, Nathan Wood, Aleeza Wilkins, Elizabeth Bondi-Kelly
7. **RELIANCE: Curating and Evaluating Reproductive Health Information on Social Media**
Vaibhav Balloli, Laura Ellis, Vishala Mishra, Alice Chi, Alex Friedman Peahl, Elizabeth Bondi-Kelly
KDD Datasets and Benchmarks, 2026 · [Project](#)
6. **PATHFinder Agent for Tailored Prenatal Care**
Vaibhav Balloli, Carissa Samuel, Samia Abdelnabi, Alex Peahl, Elizabeth Bondi-Kelly
Interactive Health, 2026 · [Project](#)
5. **“Where is this coming from?” Uncovering Trustworthiness Ideals in AI-powered Peripartum Information Seeking**
Vaibhav Balloli, Julia Erickson, Xinyi Li, Erin van Liemt, Alex Friedman Peahl, Elizabeth Bondi-Kelly
ACM FAccT, 2026
4. **SEEK-CBM: Editable and Interpretable Retrieval for Elephant Re-Identification**
Antoine Salaün, **Vaibhav Balloli**, Timm Haucke, Elizabeth Bondi-Kelly, Sara Beery
CV4Animals Workshop @ CVPR 2026 · [Talk](#)
3. **Bridging the Gap: Dynamic Learning Strategies for Improving Multilingual Performance in LLMs**
Somnath Kumar*, **Vaibhav Balloli***, Mercy Ranjit, Kabir Ahuja, Tanuja Ganu, Sunayana Sitaram, Kalika Bali, Akshay Nambi
COLING 2025 (Oral)
2. **Are They the Same Picture? Adapting Concept Bottleneck Models for Human-AI Collaboration in Image Retrieval**
Vaibhav Balloli, Sara Beery, Elizabeth Bondi-Kelly
IJCAI 2024, Human-Centered AI Track (<5% acceptance) · [Code](#) · [Project](#)
1. **Chanakya: Learning Runtime Decisions for Adaptive Real-Time Perception**
Anurag Ghosh, **Vaibhav Balloli**, Akshay Nambi, Aditya Singh, Tanuja Ganu
NeurIPS 2023 · [Talk](#) · [Code](#)

Awards and Service

- Exceptional rating, Ph.D. Preliminary Exam (advancement to candidacy) 2025
- **2nd place**, Google x MHacks Hackathon (USD 1,500); [featured by labs.google](#) 2024
- Selected for [Harvard/MIT HAIST/MAIA Intro Fellowship](#) on AI Safety 2023
- **Press**: VeLLM cited by [Satya Nadella](#) and in [Times of India](#); HAMS in [Punjab News Express](#) 2022–2024
- **Reviewer**: NeurIPS (2024–2026), KDD 2026, IJCAI 2025, ICML 2025, AAMAS 2025, ICLR 2025; workshops at CVPR 2024, NeurIPS 2023, ACL 2026
- Featured in podcast regarding SmartCampus (co-founded during undergrad) 2023